Plan of approach

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KMP
Kennis Management Platform

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1. Contents

1. Background

The organization where I am doing my internship is the HZ university of applied sciences in Vlissingen, under the subsector of the KMP (Kennis Management Platform).

1.1. Organization

The organization and company HZ university of applied sciences is a Dutch learning institution where they educate around 5000 students in the field of bachelor and master coursers. The HZ is in the Zeeland region with two different locations, Middelburg and Vlissingen.

The HZ was founded in the year 1987 due to a fusion of six different educational institutions. The first merger was between the HTS (Hogere Technische School), and the Maritiem Instituut de Ruyter, later, other institutions also were merged in the same place. The HTS was located opposite the town hall in Vlissingen, but this building lacked basic facilities such as a cafeteria. For the fusion, they needed a bigger place, so they changed location in 1967 to Edisonweg, where they are still located today. In 2011 the name HZ university of applied sciences was introduced.

Currently, HZ university of Applied Sciences holds around 4500 students and 600 people in staff (HZ University of Applied Sciences, 2024). To support the staff and students with products and services they have supporting facilities like modern learning environments, IT accessibility, study support, essential services like catering and HZ shops, global opportunities (HZ international office), and creative projects like podcast studios and AV services (*HZ University of Applied Sciences*, n.d.).
This University has an acceptance rate between 70-79%, making it a small selective institution. (*HZ University of Applied Sciences 2025 Rankings, Courses, Tuition & Admissions*, n.d.)

1.2. Mission & Vision and Core Activities

In this sub-chapter I will explain what HZs main mission and vision are. I will also talk about HZs core activities, ambitions and Plan.

1.2.1. Mission & Vision

The HZ contributes to a better world:

...by educating higher professional education professionals as the personal University of Applied Sciences;
...by, as University of Applied Sciences and together with partners, finding solutions for questions in the field of water, energy, and vitality;
...by, as a regional partner for the Zeeland Delta, supporting the developments in the region.
(HZ University of Applied Sciences, 2024).

1.2.2. Plan

To summarize HZs plan for the upcoming years (2022 – 2027), they will focus on three main strategic missions, Education, Research & Innovation, and Regional Engagement. The main idea is that students and staff will work together to tackle real-world problems related to sustainability (HZ University of Applied Sciences, 2025).

1.2.3. Education

HZ provides many bachelor's degree studies; they have got around 30 different options that are HBO-level. These bachelor’s degrees are divided into different sub-sectors such as economy, human & behavior, and technology. The university aims to prepare the students for the labor market, providing hands-on experience in combination with theoretical knowledge and practical assignments such as internships and projects (Wikipedia-bijdragers, 2024).

1.2.4. Applied Research

The HZ is not only a learning institution, but also an organization that has focus on applied research. The university collaborates closely with partner institutions, researchers, the government, and many more businesses in the Zeeland region. This ensures that the students are closely connected to the real world and get hands-on experience (HZ University of Applied Sciences, 2025).

1.3. Organizational Structure

HZ’s organogram shows how the different services and domains are structured underneath the board of directors.

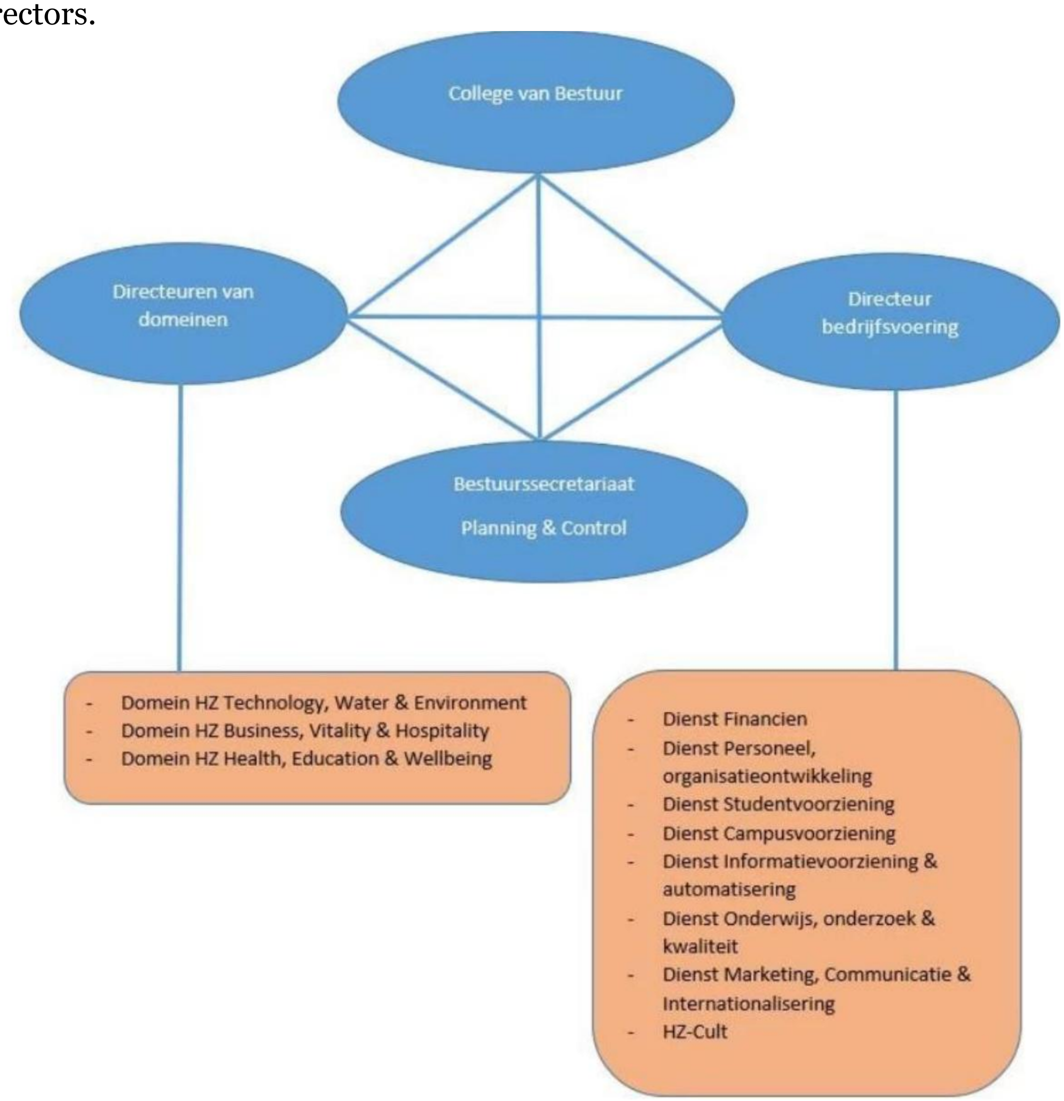

Figure 1, organogram HZ, (HZ University of Applied Sciences, 2023)

In my internship I work under the sector “Dienst Informatievoorziening & automatisering” or in English “Information Provision & Automation Service”. This sector handles ICT systems, digital platforms and knowledge management services such as the Kennis Management Platform (KMP) This enables the service of support to all study programs and staff, enabling automation tools, ICT services, and information systems that enable teaching, research, and administration.

1.4. Position of Internship

My internship is carried out within the KMP department. This department functions as a supporting unit for the whole organization, focusing on optimizing internal processes related to storing knowledge for researchers, professors, and other staff.

The main purpose of the KMP is that all knowledge that is generated within the university is stored, captured, and made accessible. When a researcher or professor finishes a project, the information must be stored so it can be reused by colleagues or applied in future research.
The KMP team makes sure that the process of maintaining knowledge goes smoothly and they make sure that the platform is managed and improved over time.

The reason I chose this internship is because of several reasons. Firstly, Opportunities for internships are limited nowadays. When I got offered an internship, I almost couldn't say no. Secondly, the project really interested me, working together with multiple departments to optimize the internal workflow for knowledge domain is something that will give me hands-on experience. Also, working with new tools like NLPs (Natural Language Processing) and LLM (Large Language Models) is something that I looked forward to, because this is used a lot within AI nowadays, which sparks my interest.

2. Problem Statement & Objective

In my internship I will be working as a data scientist for the KMP team. This team consists of 3 people, another intern and me. We work from home on Mondays and Fridays, but the office remains open for us if working from home is not an option, the rest of the week everyone is in the office. On Tuesday and Thursday, we hold small stand-ups to catch up on what everyone is doing.

Before I started the internship I did research in the field of semantic networks. This helped me in the beginning of the internship because it is a very difficult system to understand. I did this because the MediaWiki KMP is using is based on semantic networks.

2.1. Internship assignment

In this chapter I will explain the problem statement for the KMP (Kennis Management Platform) at the HZ University of Applied Sciences in relation to my internship assignment.

2.1.1. Context

The KMP is a knowledge sharing platform for research groups, (external) stakeholders, and professors to share their knowledge within the organization in combination with external parties. This knowledge can be research projects, papers, and ongoing initiatives like projects with external parties.

Within the platform, where researchers and professors can upload their projects, they are required to manually provide structured information such as summaries, titles, and metadata. This manual input can cause problems such as unstructured formatting, they can differ in language, and they can have problems with completeness.

As a result of this, the users must manually read, interpret, and translate their document content. This process could lead to repetitiveness which is time-consuming and it can increase the likelihood of inconsistent and incomplete project descriptions.

2.1.2. Problem

The core issue addressed in my internship is the lack of automated support for transforming unstructured project documents into structured consistent information that is suitable for the KMP uploading process.

Because the uploading process is performed manually, uploading projects require a significant amount of time and effort from the researchers and professors. This could lead to incomplete submissions and a reduction in platform activity. Data wise, all the valuable information remains locked in unstructured text, which makes it difficult to reuse, search, and standardize across other projects.

2.1.3. Objective & My Task

The main objective of this internship is to design and implement a Proof of Concept (PoC) that demonstrates how Natural Language Processing (NLP) techniques can support the KMP uploading process. The aim of the NLP techniques is to automatically extract structured information from different types of documents. These documents could differ in format, structure, language, and completeness. The documents include portfolios, reports, and many other different types of documents.

Making use of preprocessing- and information extraction techniques, the PoC will generate draft summaries, find key content and headings, and produce structured outputs. These outputs can be mapped to the required fields on the platform. By automating these steps, we aim to reduce the manual effort involved in uploading documents, while improving the consistency and data quality within the platform.

2.2. Learning Outcome 1

- *You can set up a data science process in a project context.*

The objective of this project is to reduce the manual workload in KMP uploading process by supporting the users with automatically generated project information. The data mining goal coming along with this is: Transforming unstructured document content into structured, reusable information that can be directly mapped to the required KMP fields (headings, summaries, key content).

To achieve this goal, I will develop a data science pipeline in which uploaded documents are transformed into JSON-based text representations. These text representations are then preprocessed through text cleaning, normalization, and segmentation. After the preprocessing steps, I will apply Natural Language Processing (NLP) techniques such as key content and heading extraction and making summaries, on the text. The output of this pipeline can be reviewed by users and used within the KMP platform.

Considering the heterogeneous nature of the data, NLP models such as BERT are most suitable for tasks such as summarization and information extraction. The choice of model is motivated by their ability to capture the context in multilingual and heterogeneous text. This is essential for the different types of documents handled by the platform.

I will divide this into SMART goals and explain it more elaborately:

SMART goals:	***Explanation:***
Specific	A data science pipeline will be developed and implemented to support the uploading process on the KMP, by automating tasks such as extracting key information from documents. The uploaded documents will be transformed into JSON-based text representations. They will also be pre-processed using pre-processing techniques and be analysed using NLP models (such as BERT) to generate summaries, headings, and key content.
Measurable	The objective is achieved when a documented data science pipeline is implemented, an unstructured document can successfully be transformed into structured JSON output, the system generates summaries, headings, and key content for test documents, and when the output is reviewed and validated by my supervisors within the KMP.
Acceptable	This objective is acceptable because it directly addresses the main problem, reducing the workload during the document uploading process. By automatically generating project information, the solution improves efficiency and accuracy while keeping the human touch through review of the generated content.
Realistic	This objective is realistic because I have access to everything I need such as the platform, documents, and the technical guidance. Furthermore, the pretrained models such as BERT can be applied locally, keeping this within company constraints, making the development of the Proof-of-Concept feasible withing the internship period.
Time-bound	The design and implementation of the pipeline will be completed within the middle phase of the internship, making it so that there is a working Proof-of-Concept in the final weeks of the internship.

To proof this is completed, I will provide a clear problem analysis, a data science pipeline, a stakeholder overview, and a validated project plan.

2.3. Learning Outcome 2

- *You can collect and address relevant data in a project context.*

Throughout this project I will learn how to collect, explore, and prepare heterogeneous data. This data will be used for analysis using NLP techniques. The documents handled in this project can contain all types of structures, unstructured, semi-structured, and elements with varying schemas. The most essential part is understanding and preparing the data for the automated information extraction.

The main goal is to collect and preprocess documents that are uploaded on the KMP and transform them into a reproducible state so that they can be used for a Proof-of-Concept of the automated information extraction system.

I will divide this into SMART goals and explain it more elaborately:

SMART goals:	***Explanation:***
Specific	I will collect representative documents that are uploaded on the KMP, including research papers and other different documents, to analyse them on their structure, language, content quality, and level of structuring. After this analysis, the documents will be transformed into a JSON-based format, where different pre-processing techniques will be applied such as text cleaning, normalization, segmentation, tokenization, and stop-word removal, to prepare them for NLP-based analysis.
Measurable	This objective will be completed when a reproducible, structured, and well-documented dataset is created. This dataset must clearly show how heterogeneous document structures are transformed into a suitable state for model testing and Proof-of-Concept development.
Acceptable	This objective is acceptable because the automated information extraction depends on consistent and well-prepared data. A proper clean dataset is necessary for the improvement of robustness, quality, and reliability of the data.
Realistic	This objective is realistic because I have access to a wide variety of documents that are uploaded on the KMP. Furthermore, I have access and knowledge from existing tools and techniques to standardize different document structures within constraints of the project.
Time-bound	This objective will start after the completion of Learning Outcome 1 once the data structure and data flow are defined. This learning outcome will be finished once the development of the models begins.

To proof I finished this learning outcome, I will provide a documented dataset, preprocessing scripts, and documentation explaining how heterogeneous document formats were prepared for automated processing.

2.4. Learning Outcome 3

- *You can perform data analysis in a project context.*

Throughout this competency, I will apply and analyze Natural Language Processing (NLP) techniques to extract information from heterogeneous document data. The focus of this competency is to analyze content, select the right modeling approaches, and evaluate the quality of the results within the Proof-of-Concept (PoC).

The main goal is to develop a PoC that automatically generates summaries, headings, key content, and metadata for the projects uploaded on the KMP.

I will divide this into SMART goals and explain it more elaborately:

SMART goals:	***Explanation:***
Specific	I will implement a PoC in which pre-processed documented data is analysed using NLP models for information extraction and summarization. This included the consideration of different NLP techniques and model configurations such as transformer-based models like BERT, to determine which is the most suitable for generating summaries, headings, and key content from heterogeneous document structures. This analysis will consider how different modelling choices affect the completeness, relevance, and usability of the generated output.
Measurable	This objective is achieved when the PoC generates useful summaries, headings, key-content, and metadata for test documents. The results will be analysed and documented, including different modelling approaches and an assessment of the quality based on predefined criteria such as relevance and useability for the uploading process on the KMP.
Acceptable	This is an acceptable objective because it demonstrates how data analysis and modelling techniques can support the automation of the uploading process on the KMP. By analysing the generated output, we can decide the suitability for different document types, this shows the technical feasibility and added value of the proposed solution.
Realistic	This objective is realistic because I have access to existing NLP frameworks and pretrained models that can be used within any technical constraint of the organization. Furthermore, I receive information and technical guidance from colleagues and other people within the organization (data science lab), that have knowledge with the platform and NLP techniques.
Time-bound	The modelling and analysis phase will be conducted after the completion of learning outcome two, after the data is prepared and processed. This phase must be completed before the final evaluation and reporting phase of my internship.

I will proof that this learning outcome is completed by showing a working proof of concept, documented, documented modelling choices, analysis of the generated outputs, and test results. This will demonstrate the quality and usability of the extracted summaries, headings, key content, and metadata.

2.5. Learning Outcome 4

- *You can evaluate and deploy results of a data science process in a project context.*

Throughout this learning outcome, I will assess the results of the data science process by analyzing the performance and usefulness of the NLP models within the PoC. The main focus is reflecting on the modeling choices, output quality, and whether the results meet the defined business and data science objectives.

The main goal of this learning outcome is to evaluate the output of the NLP-based information extraction process. Also providing a well-founded recommendation for improvement, further development, and potential deployment within the KMP itself.

I will divide this into SMART goals and explain it more elaborately:

SMART goals:	***Explanation:***
Specific	I will evaluate the quality of the results by looking at the generated summaries, headings, key content, and metadata produced by the NLP models. The evaluation is based on pre-defined criteria such as relevance, completeness, and usability within the KMP uploading process. Based on this I will evaluate the modelling choices made and formulate recommendations for improving and future integration within the KMP.
Measurable	This objective is achieved when I completed an evaluation report that documents the criteria, analysis of the model's output, the limitations, and concrete improvement proposals. This report will include whether the objectives have been met and under which conditions further development is suitable.
Acceptable	This objective is acceptable because evaluating and reflecting on the modelling decisions is essential for the practical value of the data science solution. All the insights that will be made support informed decision-making about further development and use of NLP techniques within the organization.
Realistic	The objective is realistic because the evaluation can be conducted on the potential users and supervisors using the generated outputs by the PoC. The results can be assessed by the technical and organizational constraints of the KMP platform without requiring a full deployment production system.
Time-bound	The evaluation phase will be conducted during the final weeks of the internship, after the modelling and analysis phase have been completed.

To proof this learning outcome is completed, I will provide an evaluation report. This report consists of an analysis of the model's output, reflection on the modelling decisions, and it includes recommendations for improving performance and future deployment within the KMP.

3. Citations

HZ University of Applied Sciences. (2023). *Jaarstukken HZ 2023*. https://hz.nl/uploads/documents/1.4-Over-de-HZ/1.4.2-Organisatie/Jaarverslagen/Jaarstukken-2023-versie-12-juni-2024-versie-website.pdf

HZ University of Applied Sciences. (2024, January 22). *Our vision*. https://hz.nl/en/organisation/our-vision

HZ University of Applied Sciences. (2025). *Institutional Plan of HZ University of Applied Sciences 2022 to 2027*. https://hz.nl/uploads/images/6.-Over-de-HZ/Instellingsplan/HZ-Institutional-Plan-2022-2027-Contributing-to-a-better-world-DEF.pdf

HZ University of Applied Sciences. (2025, August 20). *Organisatie*. https://hz.nl/organisatie

HZ University of Applied Sciences. (n.d.). *HZ University of Applied Sciences*. https://www.educations.com/institutions/hz-university-of-applied-sciences

HZ University of Applied Sciences 2025 Rankings, Courses, Tuition & Admissions. (n.d.). https://www.unirank.org/nl/uni/hz-university-of-applied-sciences/#google_vignette

Wikipedia-bijdragers. (2024, April 10). *HZ University of Applied Sciences*. Wikipedia. https://nl.wikipedia.org/wiki/HZ_University_of_Applied_Sciences